Discrete and Continuous Distributions

50 minutes • 50 points • Instructor key

Grading note. Award method credit after an earlier error and avoid repeated arithmetic deductions. In Problem 4, grade each step separately to distinguish modeling, geometric means, and adding stages.

Instructions. Answer all four problems and show your reasoning. No calculator is needed; exact expressions with powers, binomial coefficients, or e are acceptable. You may quote standard distribution formulas. In Problem 4, build the answer from the stages instead of simply quoting a coupon-collector formula. Geometric variables count trials *including the first success*; exponential parameters are *rates*. The suggested times leave 4 minutes for checking.

1. One trial, a count, and a wait

12 points · about 11 minutes

A device makes independent attempts to send a message. Each attempt succeeds with probability $1/3$.

(a) Let $I = 1$ if the first attempt succeeds and $I = 0$ otherwise. Name the distribution of I, including its parameter, and find $\mathbb{E}[I]$ and $\mathrm{Var}(I)$. (3 points)

Solution. $I \sim \text{Bernoulli}(1/3)$, with $\mathbb{P}(I = 1) = 1/3$ and $\mathbb{P}(I = 0) = 2/3$. Thus

$$\mathbb{E}[I] = \frac{1}{3}, \qquad \mathrm{Var}(I) = \frac{1}{3}\left(1 - \frac{1}{3}\right) = \frac{2}{9}.$$

Credit: Family and parameter: 1; mean: 1; variance: 1.

(b) Let N be the number of successes in the first six attempts. Name its distribution and parameters. Find $\mathbb{P}(N = 2)$, $\mathbb{E}[N]$, and $\mathrm{Var}(N)$. (5 points)

Solution. $N \sim \text{Bin}(6, 1/3)$ because the number of independent trials is fixed and the success probability is constant.

$$\mathbb{P}(N = 2) = \binom{6}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^4 = \frac{80}{243}, \qquad \mathbb{E}[N] = 6 \cdot \frac{1}{3} = 2, \qquad \mathrm{Var}(N) = 6 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{3}.$$

The binomial factor chooses the two successful positions.

Credit: Family and parameters: 1; probability, including the binomial factor: 2; mean: 1; variance: 1.

(c) Let T be the number of attempts up to and including the first success. Name its distribution and parameter, state its possible values, and find $\mathbb{P}(T > 3)$ and $\mathbb{E}[T]$. (4 points)

Solution. $T \sim \text{Geom}(1/3)$, with possible values $1, 2, 3, \ldots$. The event $T > 3$ means three failures, so

$$\mathbb{P}(T > 3) = \left(\frac{2}{3}\right)^3 = \frac{8}{27}, \qquad \mathbb{E}[T] = \frac{1}{1/3} = 3.$$

Credit: Family and parameter: 1; support: 1; tail probability: 1; mean: 1. A failures-before-success convention does not match T.

2. Counts and waiting times

10 points · about 9 minutes

A service receives requests according to this model: the number arriving in any interval of length t minutes has a Poisson distribution with parameter $2t$, and counts in disjoint intervals are independent. Let N be the number of requests in the next *30 seconds*, and let W be the waiting time, in *minutes*, from now until the next request.

(a) State the distribution and parameter of N, and find $\mathbb{E}[N]$ and $\mathrm{Var}(N)$. (3 points)

Solution. Thirty seconds is $1/2$ minute, so the Poisson parameter is $2(1/2) = 1$. Hence $N \sim \text{Poisson}(1)$ and $\mathbb{E}[N] = \mathrm{Var}(N) = 1$.

Credit: Distribution with correctly converted parameter: 1; mean: 1; variance: 1.

(b) Find $\mathbb{P}(N \geq 2)$. (2 points)

Solution.

$$\mathbb{P}(N \geq 2) = 1 - \mathbb{P}(N = 0) - \mathbb{P}(N = 1) = 1 - e^{-1} - e^{-1} = 1 - 2e^{-1}.$$

Credit: Correct complement or equivalent sum: 1; substituted Poisson probabilities: 1.

(c) Use the probability of *zero requests* in the next t minutes to find $\mathbb{P}(W > t)$ for $t \geq 0$. Identify the distribution of W, including its rate, and find $\mathbb{E}[W]$. (3 points)

Solution. Waiting longer than t means that the next t minutes contain no requests. Therefore

$$\mathbb{P}(W > t) = \mathbb{P}(\text{Poisson}(2t) = 0) = e^{-2t}.$$

This is $W \sim \text{Exp(rate 2)}$, supported on $[0, \infty)$, with $\mathbb{E}[W] = 1/2$ minute (30 seconds). At $t = 0$, the survival probability is 1.

Credit: Zero-count connection and survival probability: 1; exponential family and rate: 1; mean with correct units: 1.

(d) Find $\mathbb{P}(W > 3 \mid W > 2)$ and justify your answer by a calculation or the memoryless property. (2 points)

Solution. Since $\{W > 3\} \subseteq \{W > 2\}$,

$$\mathbb{P}(W > 3 \mid W > 2) = \frac{e^{-6}}{e^{-4}} = e^{-2}.$$

Equivalently, the extra wait is more than one minute with the same probability e^{-2} as a fresh exponential wait.

Credit: Conditional ratio or correctly applied memorylessness: 1; value: 1.

3. A continuous uniform wait

8 points · about 8 minutes

For a different service, the waiting time U, in minutes, has a *continuous* uniform distribution on $[0, 6]$.

(a) Write the density $f_U(u)$, including its value outside $[0, 6]$. Find $\mathbb{P}(2 < U < 5)$ and $\mathbb{P}(U = 3)$. (3 points)

Solution.

$$f_U(u) = \begin{cases} 1/6, & 0 \le u \le 6, \\ 0, & \text{otherwise.} \end{cases} \qquad \mathbb{P}(2 < U < 5) = \frac{5-2}{6} = \frac{1}{2}, \qquad \mathbb{P}(U = 3) = 0.$$

Interval probability is area under the density. A single point has zero probability, even though its density is $1/6$. Endpoint values of a density do not affect probabilities.

Credit: Density and support: 1; interval probability: 1; point probability: 1.

(b) Find $\mathbb{E}[U]$ and $\mathrm{Var}(U)$. (2 points)

Solution.

$$\mathbb{E}[U] = \frac{0+6}{2} = 3 \text{ minutes}, \qquad \mathrm{Var}(U) = \frac{(6-0)^2}{12} = 3 \text{ minutes squared}.$$

Direct integration gives the same values: $\mathbb{E}[U^2] = \int_0^6 u^2/6 \, du = 12$, so the variance is $12 - 3^2 = 3$.

Credit: Mean: 1; variance: 1.

(c) Is this uniform waiting time memoryless? Compute $\mathbb{P}(U > 5 \mid U > 2)$ and $\mathbb{P}(U > 3)$, then compare them. (3 points)

Solution.

$$\mathbb{P}(U > 5 \mid U > 2) = \frac{\mathbb{P}(U > 5)}{\mathbb{P}(U > 2)} = \frac{1/6}{4/6} = \frac{1}{4}, \qquad \mathbb{P}(U > 3) = \frac{3}{6} = \frac{1}{2}.$$

These are unequal, so U is not memoryless. After two minutes have elapsed, the remaining wait is at most four minutes; its distribution has changed.

Credit: Conditional probability: 1; unconditional tail: 1; conclusion from their inequality: 1.

4. Coupon collection, one stage at a time

20 points · about 18 minutes

Each pack contains one sticker, equally likely to be of type A, B, C, or D. Packs are independent; duplicates are possible. You start with no stickers. Let C be the number of packs needed to collect all four types. After exactly k distinct types have been collected, let W_k count *additional packs up to and including the next new type*, for $k = 0, 1, 2, 3$.

You may use: if $G \sim \text{Geom}(p)$ counts trials through the first success, then $\mathbb{E}[G] = 1/p$.

(a) Your first five stickers are A, A, C, A, C. Which types would make the next sticker new, and what is the probability it is new? (2 points)

Solution. Only B and D are new, so the probability is $2/4 = 1/2$. The relevant count is two distinct types, not five packs.

Credit: Missing types: 1; probability: 1.

(b) After k distinct types have been collected, find the probability p_k that the next pack gives a new type. Why does this probability stay the same when a duplicate appears? (3 points)

Solution. There are $4 - k$ missing types out of four equally likely types, so $p_k = (4 - k)/4$. A duplicate does not change the set of collected types, and the next pack is still independent and uniform.

Credit: Number missing: 1; success probability: 1; unchanged collection and draw law after a duplicate: 1.

(c) State the distribution of W_k for $k = 1, 2, 3$. Fill in the table, treating W_0 directly. (5 points)

Distinct types k	New-type probability p_k	Expected wait $\mathbb{E}[W_k]$
0	1	1
1	3/4	4/3
2	1/2	2
3	1/4	4

Solution. Within each stage the independent draws have fixed success probability p_k, so $W_k \sim \text{Geom}(p_k)$ and $\mathbb{E}[W_k] = 1/p_k$. The first pack is certainly new, so $W_0 = 1$.

Credit: Geometric identification with the stage parameter: 1; each complete table row: 1 (half per correct entry). Accept Geom(1) for the deterministic W_0.

(d) Express C in terms of W_0, W_1, W_2, W_3 and compute $\mathbb{E}[C]$. Name the expectation rule you use. Does that rule require the stages to be independent? (4 points)

Solution. $C = W_0 + W_1 + W_2 + W_3$. By linearity of expectation, $\mathbb{E}[C] = 1 + 4/3 + 2 + 4 = 25/3$ packs. Linearity does not require independence of the stages.

Credit: Decomposition: 1; linearity: 1; sum and value: 1; no independence requirement: 1. A quoted $4H_4$ alone earns only the numerical-result point.

(e) Suppose you currently have exactly two distinct types. What is the expected number of *additional* packs to finish? Explain why $4/2 = 2$ is not the answer to this question. (3 points)

Solution. Two stages remain, so the expected additional wait is $\mathbb{E}[W_2] + \mathbb{E}[W_3] = 2 + 4 = 6$ packs. The value $4/2 = 2$ is only the expected wait for the next new type; after that, the final missing type has success probability $1/4$.

Credit: Two remaining stages: 1; sum 6: 1; explain what the value 2 actually measures: 1.

(f) A student claims that C itself has a geometric distribution with parameter $1/4$. Is this correct? Give a reason using the possible values of C or the changes between stages. (3 points)

Solution. No. At least four packs are needed, so $\mathbb{P}(C = 1) = 0$, but a Geom(1/4) variable equals 1 with probability $1/4$. Alternatively, each stage waits for one new type with a different success probability; completion requires all four stages, not one fixed-probability first success.

Credit: Reject the claim: 1; a valid support comparison or explanation of the changing stages: 2.